Practical -14

Packet sniffing

AIM:

To write code using raw sockets to implement packet sniffing.

CODE:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "unknown protocol"

        print(f"Protocol: {protocol_name}")
        print(f"Source ip: {src_ip}")
        print(f"Destination ip: {dst_ip}")
        print("-" * 50)

sniff(iface="WiFi", prn=packet_callback, filter="ip",
      store=0)
```

OUTPUT:

PROTOCOL : TCP
SOURCE IP : 204.79.197.254
DESTINATION IP : 192.168.1.7

PROTOCOL : TCP
SOURCE IP : 192.168.1.7
DESTINATION IP : 148.113.9.134

PROTOCOL : TCP
SOURCE IP : 148.113.9.134
DESTINATION IP : 192.168.1.7

PROTOCOL : UDP
SOURCE IP : 192.168.1.7
DESTINATION IP : 192.168.1.255

PROTOCOL : UDP
SOURCE IP : 192.168.1.7
DESTINATION IP : 192.168.1.255

RESULT:

Thus the packet sniffing program was successfully implemented to capture and display packet details.